



北京邮电大学



For examiners' use only

BBU4208 B

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Total	

Joint Programme Examinations 2022/23

BBU4208 Data Structure

Paper B

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number

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BUPT student number

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Class number

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NOT allowed: electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. You must **NOT** take answer books, used or unused, from the examination room.
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Zhang Xi, Dr Xu Ke, Dr Guo Sanchuan

Instructions

Before the start of the examination

- 1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.
- 2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance. Possession of mobile phones, electronic devices and unauthorised materials is an offence.**
- 3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**
- 4) Do not turn over your question paper or begin writing until told to do.

During the examination

- 1) You must not communicate with or copy from another student.
- 2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.
- 3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.
- 4) For 2 hour examinations you may **not** leave temporarily.
- 5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

- 1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**
- 2) Remain in your seat until you are told you may leave.

Question 1**[30 marks, 2 marks for each]**

1. If each data element in a data structure may have only one immediate predecessor, but multiple immediate successors, then the data structure is
A. Stack B. Queue
C. Tree D. Graph

2. Assuming that the head pointer of a single linked list with a head node is *head*, the condition for determining that the table is an empty table is
A. $head == \text{NULL}$ B. $head \rightarrow next == \text{NULL}$
C. $head != \text{NULL}$ D. $head \rightarrow next == head$

3. The space complexity of heap-sorting a key sequence of length n is
A. $O(\log_2 n)$ B. $O(1)$
C. $O(n)$ D. $O(n * \log_2 n)$

4. Suppose the elements of the circular queue are stored in the array $A[n]$. The *front* pointer is set to point to the front element in the queue, and the *rear* pointer points to the next position of the rear element in the queue. The number of elements currently stored in the queue is
A. $(rear - front - 1) \% n$ B. $(rear - front) \% n$
C. $(front - rear + 1) \% n$ D. $(rear - front + n) \% n$

5. The maximum size of a stack is 4. If the push sequence is 1, 2, 3, 4, 5, 6, and the push and pop can be interspersed, the possible pop sequence is
A. 5, 4, 3, 2, 1, 6 B. 2, 3, 5, 6, 1, 4
C. 3, 2, 5, 4, 1, 6 D. 1, 4, 6, 5, 2, 3

6. It is known that there is only one leaf node in a binary tree with 50 nodes, then the number of nodes with degree 1 in the tree is
A. 0 B. 1
C. 48 D. 49

7. Assuming that the directed graph contains n vertices and e arcs, it means that the number of arc nodes contained in the adjacency list of the graph is
A. n B. e
C. $2e$ D. $n * e$

8. It is known that the inorder sequence and postorder sequence of a binary tree are both ABCDEF, then the preorder sequence of the binary tree is
A. FEDCBA B. ABCDEF
C. FDECBA D. FBDCEA

9. If an undirected graph G contains 21 edges, then the number of vertices in G is at least
A. 7 B. 8
C. 21 D. 22

10. It is known that in a tree of degree 3, the number of nodes of degree 2 is 4, and the number of nodes of degree 3 is 3, then the number of leaf nodes in the tree is
A. 5 B. 8
C. 11 D. 18

11. The number of different topological sequences that can be obtained by the directed acyclic graph as shown in Figure 1 is

- A.1
C.3
B.2
D.4

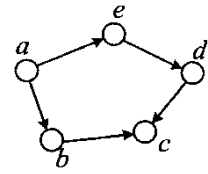


Figure 1

12. It is known that the adjacency matrix of an undirected graph containing 6 vertices ($v_0, v_1, v_2, v_3, v_4, v_5$) is shown in Figure 2, then the vertex access sequence that may be obtained by depth-first traversal from vertex v_0 is:

- A. ($v_0, v_1, v_2, v_5, v_4, v_3$)
B. ($v_0, v_1, v_2, v_3, v_4, v_5$)
C. ($v_0, v_1, v_5, v_2, v_3, v_4$)
D. ($v_0, v_1, v_4, v_5, v_2, v_3$)

		0	1	2	3	4	5
0	V_0	0	1	1	0	0	0
1	V_1	1	0	1	1	0	0
2	V_2	1	1	0	0	0	1
3	V_3	0	1	0	0	0	0
4	V_4	0	0	0	0	0	1
5	V_5	0	0	1	0	1	0

Figure 2

13. Among the following sorting methods, the stable sorting method is

- A. Shell sort
C. Quick sort
B. Heap sort
D. Insertion sort

14. For stable sorting in $O(n \cdot \log_2 n)$ time complexity, the available sorting methods is

- A. Merge sort
C. Heap sort
B. Quick sort
D. Bubble Sort

15. The storage space of the known hash table is $T[0..18]$, the hash function $H(\text{key}) = \text{key} \% 17$, and the collision is handled by linear probing mechanism. The following keywords have been inserted into the hash table: $T[5] = 39$, $T[6] = 57$ and $T[7] = 7$, then the position where the next keyword 23 is inserted is

- A. $T[2]$
C. $T[8]$
B. $T[4]$
D. $T[10]$

		Do not write in this column
1)	2)	
3)	4)	
5)	6)	
7)	8)	
9)	10)	
11)	12)	
13)	14)	
15)		30 marks

Question 2 Filling the blanks**[20 marks, 2 marks for each]**

1. If we establish a singly linked list with a head node, and the data are n elements from a linear list, the time complexity is_____.
2. If we merge two increasing ordered singly linked lists of lengths m and n into a singly linked list with elements in descending order, the best time complexity that can be achieved is_____.
3. Assuming that the storage space for a circular queue is m elements. The pointer *front* points to the element in the front of the queue, and the pointer *rear* points to the next position of the element in the rear of the queue. To check whether the queue is full, we should use the logic expression_____.
4. Assuming that we have defined a function which performs string concatenation:

```
char *strcat(char *to, char *from);
```

// joins the string 'from' to the end of the string 'to' and returns the joined string
If the string $s1 = \text{"point"}$, $s2 = \text{"of"}$, then the value returned by `strcat(s1, strcat(s2, s1))` is_____.
5. Among the binary trees which has 3 nodes named a , b and c respectively, there are in total _____ different binary trees for which the preorder sequence is "abc" and the postorder sequence is "cba".
6. Given a directed graph as shown in Figure 3, the length of the shortest path from vertex 1 to vertex 4 is _____.

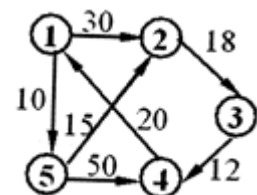


Figure 3

7. If the directed graph is represented by an adjacency matrix, then the in-degree of vertex i is equal to the _____ in the matrix.
8. Among quicksort, heapsort and mergesort, _____ has the worst time complexity of $O(n^2)$.
9. Compared with binary search, there are two advantages of sequential search. One advantage is that the data elements in the table are not needed to be ordered, and the other advantage is that there is no special requirement for_____.
10. Assuming that the length of the hash table is 11, and the hash function is $H(\text{key}) = \text{key} \% 7$. If linear probing is used to deal with conflicts, the calculation formula of the probing address sequence h_i is _____.

		Do not write in this column
1)	2)	
3)	4)	
5)	6)	
7)	8)	
9)	10)	20 marks

Question 3 Answer the Questions

[30 marks, 6 marks for each]

1. The adjacency list of a directed graph is shown in Figure 4.
 - (1) Write the vertex sequence for breadth-first search traversal of the graph starting from vertex A.
 - (2) Draw the inverse adjacency list of the directed graph.

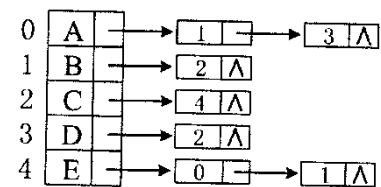


Figure 4

[illegible]

2. Assumes that two stacks use a shared storage space `data[m]`, which are defined by the code as follows:

```
typedef struct {
    DataType data[m];
    int top1, top2;
} SeqStack;
```

The bottoms of the two stacks are set to the two sides of data[m]. Initially, top1=-1, top2=m.

- (1) Design a function which checks whether the stack is full: `int stackfull(SeqStack *s);`
- (2) Design a push function for the stack: `void push(SeqStack *s, int si, DataType x).` When `si==0`, the element is pushed into the stack with the bottom `data[0]`, and when `si==1`, the element is pushed into the stack with the bottom `data[m-1]`.

[illegible]

5. For a given sequence {26, 18, 60, 65, 45, 13, 32}, write the procedure for sorting them into an ascending sequence using the selection sort algorithm.

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	6 marks

Question marking: $\frac{\quad}{6} + \frac{\quad}{6} + \frac{\quad}{6} + \frac{\quad}{6} + \frac{\quad}{6} = \frac{\quad}{30}$

Question 4 Algorithm Design

[20 marks, 10 marks for each]

1. Assuming that a linear list is represented by a singly linked list, the type of a singly linked list is defined as follows:

```
typedef struct node {
    DataType data;
    struct node *next;
} LinkNode, *LinkList;
```

Write an algorithm, transform a singly linked list without the head node into a singly linked list with a head node, and then analyze the time complexity of the algorithm.

[illegible]

2. The type of a binary search tree is defined as follows:

```
typedef struct BSTNode {
    int data;
    struct BSTNode *lchild, *rchild;
}BSTNode,*BSTree;
```

Design a recursive algorithm to count the number of nodes in the binary search tree, whose value is less than a given value a .

[illegible]

Question marking: $\frac{\quad}{10} + \frac{\quad}{10} = \frac{\quad}{20}$

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